

b) the function that assigns to each positive integer the number of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that do not appear as decimal digits of the integer.

Solution: Domain: $\mathbb{Z}^+$

There are 10 different digits. For a positive integer having x different digits where $1 \leq x \leq 10$, the function returns $10-x$.

$\therefore 10-1 \geq 10-x \geq 10-10 \Rightarrow 0 \leq x \leq 9$. Range: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

c) the function that assigns to a bit string the number of times the block 11 appears.

Solution: Domain: The set of all possible bit strings.

Range: $\mathbb{N}$

d) the function that assigns to a bit string the numerical position of the first 1 in the string and that assigns the value 0 to a bit string consisting of all 0s.

Solution: Domain: The set of all possible bit strings

Range: $\mathbb{N}$

9) Find these values.

a) $\lceil \frac{3}{4} \rceil \rightarrow 1$ b) $\lfloor \frac{7}{8} \rfloor \rightarrow 0$ c) $\lceil -\frac{3}{4} \rceil \rightarrow 0$ d) $\lfloor -\frac{7}{8} \rfloor \rightarrow -1$
 e) $\lceil 3 \rceil \rightarrow 3$ f) $\lfloor -1 \rfloor \rightarrow -1$ g) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor \rightarrow \lfloor \frac{1}{2} + 2 \rfloor \rightarrow 2$
 h) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor = \lfloor \frac{1}{2} \cdot 2 \rfloor = 1$

12) Determine whether each of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one?

a) $f(n) = n-1$

Solution: Let $x, y \in \mathbb{Z}$ s.t. $f(x) = f(y)$. $\therefore x-1 = y-1 \Rightarrow x = y$.

$\therefore$ By Defn. 5, f is one-to-one.

b) $f(n) = n^2 + 1$

Solution: $f(3) = 3^2 + 1 = 10$ $f(-3) = (-3)^2 + 1 = 10$. But $3 \neq -3$.
 $\therefore$ By Defn. 5, f is not one-to-one.

c) $f(n) = n^3$

Solution: Let $x, y \in \mathbb{Z}$ s.t. $f(x) = f(y)$. $\therefore x^3 = y^3 \Rightarrow x^3 - y^3 = 0$
 $\Rightarrow (x-y)(x^2 + xy + y^2) = 0$. Either $(x-y) = 0$ or $x^2 + xy + y^2 = 0$